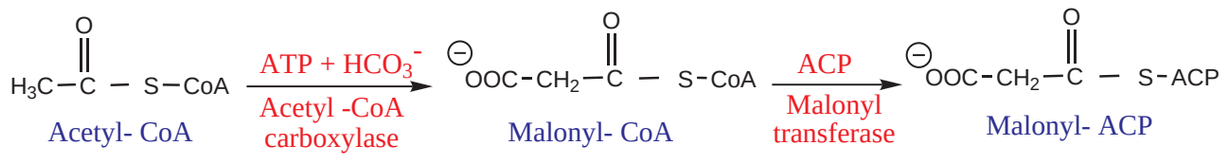
Within the mitochondria, the acetyl CoA reacts with oxaloacetate in a reaction catalyzed by citrate synthase to form citrate. The acetyl group passes out of the mitochondria as citrate. In the cytosol, the citrate reacts with cytosolic CoA to form acetyl CoA and oxaloacetate. Oxaloacetate is reduced to malate and then transported to mitochondrial matrix.

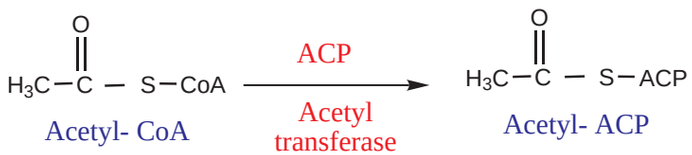
Fatty acid synthesis takes place in the cytosol in two steps.

1. Synthesis of palmitic acid
2. Elongation of fatty acid

Synthesis of palmitic acid:

Palmitic acid is a 16-carbon saturated fatty acid. A sequence of 7 enzyme catalyzed reactions converts 2 carbons unit to 4 carbons unit. First three steps of biosynthesis involve formation of acetyl ACP and malonyl ACP from acetyl CoA.





The last four enzymatic reactions of fatty acid biosynthesis involve the elongation step which includes condensation, reduction, dehydration and reduction. Repetition of this elongation step, increases the chain length by 2 carbon atoms. 7 elongation steps produce palmitic acid.

Elongation of fatty acids:

The normal product of fatty acid biosynthesis at cytosol is palmitic acid. Fatty acids longer than 16 C are formed through the addition of extra 2 carbon units by elongation reaction. The fatty acids elongation happens in both mitochondria and endoplasmic reticulum.

5.3 OXIDATION OF FATTY ACIDS

The digestion of fats starts in the small intestine. Fats are emulsified by bile salts and hydrolyzed by pancreatic lipases to form free fatty acids. These free fatty acids combine with glycerol phosphate (derived from dihydroxy acetone phosphate produced by the glycolytic process) to form triglycerides. They combine with proteins to form lipoproteins and enter circulation to perform various biological functions such as oxidation, storage and formation of new lipids. Thus, the various fatty acids may exist in the free form as well

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